

Boss Challenge — Paper 07 (Class 7)

Winds, Storms & Cyclones | Wastewater Story | The Triangle & its Properties

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Moving air is called:
(A) rain
(B) sound
(C) heat
(D) wind
2. Water that carries human and household wastes (used water from kitchens, toilets, baths) is called:
(A) river water
(B) sewage (wastewater)
(C) distilled water
(D) rainwater
3. The three angles of any triangle always add up to:
(A) 180°
(B) 100°
(C) 90°
(D) 360°
4. Wind blows mainly because air moves from a region of:
(A) up to down only
(B) low pressure to high pressure
(C) cold to hot only
(D) high pressure to low pressure
5. The FIRST step in cleaning sewage at a treatment plant is to:
(A) freeze it
(B) pass it through bar screens to remove large floating objects
(C) add sugar
(D) drink it
6. Two angles of a triangle are 50° and 60° . The third angle is:
(A) 60°
(B) 110°
(C) 90°
(D) 70°
7. Warm air rises because it is:
(A) heavier than cool air
(B) colder
(C) solid
(D) lighter (less dense) than cool air

8. In a treatment plant, air is bubbled through the water in the aeration tank so that:
- (A) large stones are removed
 - (B) the water freezes
 - (C) the water becomes sweet
 - (D) helpful bacteria can grow and digest the wastes
9. A triangle with all three sides equal is called:
- (A) equilateral
 - (B) isosceles
 - (C) scalene
 - (D) right
10. The instrument used to measure wind speed is the:
- (A) anemometer
 - (B) thermometer
 - (C) barometer
 - (D) rain gauge
11. Why should we NOT pour cooking oil and fats down the kitchen drain?
- (A) they clean the pipes
 - (B) they remove germs
 - (C) they make water sweet
 - (D) they harden and block the drains/pipes
12. Each angle of an equilateral triangle measures:
- (A) 180°
 - (B) 60°
 - (C) 90°
 - (D) 45°
13. Air all around us pushes on things. This shows that air:
- (A) exerts pressure
 - (B) has no weight
 - (C) cannot move
 - (D) is solid
14. Cleaned water from a treatment plant is usually:
- (A) thrown into the air
 - (B) released back into rivers or the ground
 - (C) bottled for drinking
 - (D) burnt
15. A triangle with one angle equal to 90° is called a:
- (A) acute
 - (B) equilateral
 - (C) obtuse
 - (D) right-angled triangle

16. The calm, clear region at the centre of a cyclone is called the:

- (A) tail
- (B) storm wall only
- (C) eye
- (D) base

17. Sewage is harmful mainly because it contains:

- (A) disease-causing germs and harmful wastes
- (B) only clean salt
- (C) pure oxygen
- (D) nothing harmful

18. Two sides of a triangle are equal. It is an:

- (A) isosceles triangle
- (B) equilateral
- (C) scalene
- (D) right

19. Uneven heating of the equator and the poles by the Sun is a main cause of:

- (A) winds blowing around the Earth
- (B) earthquakes
- (C) tides
- (D) eclipses

20. Large solid objects and small stones are removed from sewage in the:

- (A) river itself
- (B) drinking-water tap
- (C) bar screens and grit/sand tank
- (D) aeration tank

21. The solid material that settles at the bottom of the sedimentation tank... (wait, that's wastewater).

The line joining a vertex of a triangle to the midpoint of the opposite side is the:

- (A) altitude
- (B) base
- (C) hypotenuse
- (D) median

22. When wind speed increases, the air pressure it exerts:

- (A) increases
- (B) decreases
- (C) stays the same
- (D) becomes zero

23. The solid waste that settles at the bottom of the settling tank during treatment is called:

- (A) sludge
- (B) clear water
- (C) grit
- (D) oxygen

24. In a right-angled triangle the longest side, opposite the right angle, is the:

- (A) base
- (B) hypotenuse
- (C) altitude
- (D) median

25. A thunderstorm usually forms when:

- (A) there is no moisture
- (B) the ground freezes
- (C) warm, moist air rises rapidly and cools to form towering clouds
- (D) cold dry air sinks slowly

26. Which of these should NOT be thrown into the drain?

- (A) nothing ever
- (B) used cooking oil, tea leaves and chemicals
- (C) rainwater
- (D) only plain water

27. Can a triangle have sides 3 cm, 4 cm and 8 cm?

- (A) Yes, because 8 is the longest
- (B) No, because all sides differ
- (C) No, because $3 + 4$ is not greater than 8
- (D) Yes, any three lengths work

28. Why is a cyclone more likely to form over a warm sea?

- (A) warm water heats the air, which rises and creates very low pressure
- (B) deserts make cyclones
- (C) cold water makes cyclones
- (D) ice makes cyclones

29. Poor sanitation and untreated sewage spread diseases such as:

- (A) cholera and typhoid
- (B) a broken bone
- (C) short-sightedness
- (D) colour-blindness

30. The exterior angle of a triangle equals:

- (A) 90° always
- (B) 180°
- (C) the third interior angle
- (D) the sum of the two opposite interior angles

31. To stay safe during a thunderstorm with lightning, you should:
- (A) lie in an open field
 - (B) stay indoors and avoid sheltering under a tall tree
 - (C) stand under the tallest tree
 - (D) hold a metal rod up high
32. Sludge from a treatment plant can be usefully turned into:
- (A) glass
 - (B) petrol
 - (C) drinking water
 - (D) biogas (fuel) or manure
33. A triangle in which all angles are less than 90° is called:
- (A) acute-angled
 - (B) right-angled
 - (C) obtuse-angled
 - (D) straight
34. Two balloons hang close together. You blow hard between them. They will:
- (A) not move
 - (B) burst
 - (C) move closer together
 - (D) move apart
35. An improved, water-saving toilet that uses earthworms to treat waste is a:
- (A) flush-only toilet
 - (B) vermi-processing toilet
 - (C) septic only
 - (D) open drain
36. Two angles of a triangle are 90° and 30° . The third angle is:
- (A) 120°
 - (B) 60°
 - (C) 30°
 - (D) 90°
37. Storms and cyclones cause damage mainly through:
- (A) snowfall
 - (B) gentle breeze
 - (C) very strong winds, heavy rain and flooding
 - (D) bright sunshine

38. Before sewage reaches a treatment plant, it is carried through a network of underground pipes called:
- (A) sewers
 - (B) canals for farming
 - (C) water mains for drinking
 - (D) rivers
39. Can the angles of a triangle be 100° , 50° and 40° ?
- (A) No, because they add up to 190° , not 180°
 - (B) No, because they are all different
 - (C) Yes, any angles work
 - (D) Yes, because one is obtuse
40. A barometer is an instrument used to measure:
- (A) rainfall
 - (B) air pressure
 - (C) wind speed
 - (D) temperature
41. Why is it important to treat sewage before letting it into a river?
- (A) to remove germs and wastes so the river is not polluted
 - (B) to colour the water
 - (C) to make the river deeper
 - (D) to slow the river down
42. The shortest distance from a vertex of a triangle, drawn perpendicular to the opposite side, is the:
- (A) base
 - (B) median
 - (C) hypotenuse
 - (D) altitude (height)
43. Monsoon winds that bring rain to India are caused mainly by:
- (A) earthquakes
 - (B) uneven heating of land and the sea
 - (C) volcanoes
 - (D) the Moon's pull
44. Which is a good habit for better sanitation?
- (A) using a proper toilet and washing hands with soap
 - (B) dumping oil in drains
 - (C) defecating in the open
 - (D) throwing waste in ponds

45. In an isosceles triangle, the two angles opposite the equal sides are:
- (A) always 90°
 - (B) always 60°
 - (C) always different
 - (D) equal to each other
46. High-speed winds can lift the roof off a house during a storm because:
- (A) the roof becomes a magnet
 - (B) fast wind above the roof lowers the pressure, and higher pressure inside pushes the roof up
 - (C) rain glues the roof up
 - (D) the wind is heavier than the roof
47. The MAIN aim of a sewage treatment plant is to:
- (A) make drinking water for homes
 - (B) produce electricity for the city
 - (C) clean dirty water enough to return it safely to nature
 - (D) store rainwater
48. If the angles of a triangle are in the ratio 1 : 2 : 3, the angles are:
- (A) 20° , 40° , 60°
 - (B) 40° , 60° , 80°
 - (C) 60° , 60° , 60°
 - (D) 30° , 60° , 90°
49. Cyclone alerts and warnings are important because they:
- (A) are just for fun
 - (B) give people time to move to safety before the cyclone hits
 - (C) stop the rain
 - (D) make the cyclone stronger
50. Contaminants in sewage include organic wastes, harmful microbes and:
- (A) nothing else
 - (B) pure oxygen
 - (C) dissolved nutrients/chemicals like nitrates and phosphates
 - (D) clean salt only
51. Can a triangle have TWO right angles?
- (A) Yes, always
 - (B) Only if equilateral
 - (C) No, because two right angles already total 180° , leaving nothing for the third
 - (D) Yes, if it is large
52. Which best explains why a cyclone weakens after it moves onto land?
- (A) the Sun switches off
 - (B) land makes it stronger
 - (C) it loses its supply of warm, moist sea air
 - (D) it freezes instantly

53. Which is the correct order of sewage treatment?
- (A) drinking -> screening -> bottling
 - (B) aeration -> screening -> settling
 - (C) settling -> drinking -> grit
 - (D) screening -> grit removal -> settling (sludge) -> aeration -> cleaner water
54. A right triangle has legs 3 cm and 4 cm. The hypotenuse is:
- (A) 7 cm
 - (B) 12 cm
 - (C) 5 cm
 - (D) 25 cm
55. Air pressure is generally LOWER at the centre of a cyclone. This is why:
- (A) there is no wind
 - (B) winds blow outward only
 - (C) winds rush inward towards the low-pressure eye
 - (D) pressure is highest there
56. Open drains and pools of stagnant dirty water are unhealthy because they:
- (A) are harmless
 - (B) give safe drinking water
 - (C) clean the air
 - (D) become breeding places for mosquitoes and spread disease
57. If two angles of a triangle are equal and each is 70° , the third angle is:
- (A) 70°
 - (B) 20°
 - (C) 40°
 - (D) 140°
58. Which is a sensible action when a cyclone warning is issued for your coastal town?
- (A) move to a safe shelter and store drinking water and food
 - (B) fly a kite
 - (C) go to the beach to watch
 - (D) ignore the warning
59. Why should factories treat their wastewater before releasing it?
- (A) it is already clean
 - (B) treatment makes it dirtier
 - (C) it may contain poisons and chemicals that harm rivers and life
 - (D) rivers like chemicals
60. The sum of the lengths of any two sides of a triangle is always:
- (A) less than the third side
 - (B) equal to 180°
 - (C) greater than the third side
 - (D) equal to the third side

61. Which list is in order of INCREASING wind strength?

- (A) breeze -> cyclone -> wind
- (B) strong wind -> cyclone -> breeze
- (C) cyclone -> breeze -> wind
- (D) gentle breeze -> strong wind -> cyclone

62. A simple thing each of us can do to help is:

- (A) throwing food in drains
- (B) pouring paint down the sink
- (C) not throwing solid waste, oil or chemicals into drains
- (D) flushing plastic

63. A triangle with all three sides of different lengths is called:

- (A) equilateral
- (B) isosceles
- (C) right
- (D) scalene

64. Why does a tightly closed empty plastic bottle get crushed a little when air is sucked out?

- (A) water fills it
- (B) outside air pressure pushes in when inside pressure drops
- (C) plastic shrinks by itself
- (D) the bottle gains weight

65. Aerobic bacteria in the aeration tank are called helpful because they:

- (A) freeze the water
- (B) feed on and break down the organic wastes in the water
- (C) make the water poisonous
- (D) add dirt to the water

66. Find the third angle of a right triangle whose one acute angle is 35° .

- (A) 35°
- (B) 65°
- (C) 55°
- (D) 145°

67. Cyclones are given different local names. In the Indian Ocean region a severe cyclonic storm is broadly a type of:

- (A) a dust devil
- (B) tropical cyclone
- (C) an earthquake
- (D) a gentle land breeze

68. If a city had NO sewage treatment, the most likely result would be:
- (A) no effect at all
 - (B) more rainfall
 - (C) polluted rivers and more water-borne diseases
 - (D) cleaner rivers
69. The area of a triangle with base 10 cm and height 6 cm is:
- (A) 16 cm²
 - (B) 30 cm²
 - (C) 60 cm²
 - (D) 30 cm
70. A 'lightning conductor' on a tall building works by:
- (A) stopping the rain
 - (B) safely carrying the lightning's electricity down to the ground
 - (C) attracting more lightning to harm people
 - (D) making thunder louder
71. Which is the cleanest, safest source among these?
- (A) untreated sewage
 - (B) a stagnant roadside pool
 - (C) treated tap water from a proper supply
 - (D) water from an open drain
72. Which set of side lengths CAN form a triangle?
- (A) 4 cm, 4 cm, 9 cm
 - (B) 6 cm, 8 cm, 10 cm
 - (C) 1 cm, 1 cm, 5 cm
 - (D) 2 cm, 3 cm, 7 cm
73. Air pressure decreases as you go higher up a mountain. This is why:
- (A) it becomes harder to breathe at very high altitudes
 - (B) there is more oxygen
 - (C) the air gets heavier
 - (D) gravity disappears
74. Besides building treatment plants, an important way to reduce the sewage problem is to:
- (A) use less water and avoid putting waste/chemicals into drains
 - (B) ignore leaks
 - (C) pour everything down the sink
 - (D) waste as much water as possible

75. The angle-sum property helps us find a missing angle. In a triangle with angles x , x and 80° , the value of x is:
- (A) 80°
 - (B) 100°
 - (C) 40°
 - (D) 50°
76. Which causes a sea breeze in the daytime (a small-scale wind)?
- (A) rainfall only
 - (B) the sea heating faster
 - (C) the Moon's gravity
 - (D) land heating faster than the sea, so cool sea air moves in
77. 'Sanitation' mainly means:
- (A) keeping our surroundings clean and disposing of human waste safely
 - (B) making electricity
 - (C) growing more crops
 - (D) building tall houses
78. The longest side of a triangle is always opposite the:
- (A) largest angle
 - (B) right angle only
 - (C) middle angle
 - (D) smallest angle
79. Which statement about air pressure and wind is TRUE?
- (A) wind blows low to high pressure
 - (B) air has no pressure
 - (C) faster air has higher pressure
 - (D) faster-moving air has lower pressure, and wind blows from high to low pressure
80. Which is the BEST single reason to treat sewage?
- (A) to create more sludge
 - (B) to protect human health and the environment
 - (C) to use up extra electricity
 - (D) to make rivers look blue
81. If one exterior angle of a triangle is 120° , the sum of its two opposite interior angles is:
- (A) 120°
 - (B) 240°
 - (C) 60°
 - (D) 180°

- 82.** Cyclones, hurricanes and typhoons are:
- (A) the same kind of storm with different names in different regions
 - (B) completely different things
 - (C) kinds of rainbows
 - (D) types of earthquakes
- 83.** After full treatment, the water released is:
- (A) much cleaner but usually still not for drinking
 - (B) 100% pure drinking water
 - (C) unchanged
 - (D) dirtier than before
- 84.** Two angles of a triangle add up to 100° . The third angle is:
- (A) 260°
 - (B) 100°
 - (C) 90°
 - (D) 80°
- 85.** The way strong storm winds knock down trees and signs shows that moving air can:
- (A) become solid
 - (B) push things with great force
 - (C) pass through solid walls
 - (D) disappear
- 86.** Decomposed sludge from a treatment plant is useful as:
- (A) drinking water
 - (B) window glass
 - (C) petrol for cars
 - (D) manure for farms
- 87.** A triangle with angles 45° , 45° and 90° is a:
- (A) scalene
 - (B) obtuse
 - (C) right-angled isosceles triangle
 - (D) equilateral
- 88.** When a cyclone warning is given, people on the coast should:
- (A) go fishing in the sea
 - (B) climb tall coconut trees
 - (C) move inland to safe shelters
 - (D) stand on the beach to watch
- 89.** Solid waste like plastic and cloth should be kept out of toilets and drains because it:
- (A) makes water sweet
 - (B) cleans the pipes
 - (C) blocks the sewer pipes
 - (D) adds oxygen

90. The three medians of any triangle:
- (A) meet at a single point
 - (B) never meet
 - (C) meet outside the triangle
 - (D) form a square
91. Which statement about wind is TRUE?
- (A) wind flows from low to high pressure
 - (B) wind is solid
 - (C) wind has no effect
 - (D) wind is caused by differences in air pressure
92. Open defecation (going outside in the open) is harmful because it:
- (A) is hygienic
 - (B) treats sewage
 - (C) saves water
 - (D) pollutes soil and water and spreads disease
93. A right triangle has legs 5 cm and 12 cm. The hypotenuse is:
- (A) 7 cm
 - (B) 13 cm
 - (C) 169 cm
 - (D) 17 cm
94. A short, violent storm with lightning, thunder and heavy rain is called a:
- (A) dew
 - (B) thunderstorm
 - (C) fog
 - (D) gentle breeze
95. Washing hands with soap before eating mainly helps to:
- (A) make food tastier
 - (B) add vitamins
 - (C) remove germs and prevent disease
 - (D) cool the food
96. An equilateral triangle (all angles 60°) is also:
- (A) obtuse-angled
 - (B) scalene
 - (C) right-angled
 - (D) an acute-angled triangle
97. Why are cyclones common along coasts but rare in inland deserts?
- (A) inland areas have more trees
 - (B) cyclones form over warm seas, which lie near coasts
 - (C) deserts are too hot for any storm
 - (D) deserts always have low pressure

98. Dirty water from homes is carried to the treatment plant through:

- (A) underground sewer pipes
- (B) open rivers
- (C) irrigation canals
- (D) rain clouds

99. Two angles of a triangle are equal and the third is 100° . Each equal angle is:

- (A) 80°
- (B) 100°
- (C) 40°
- (D) 50°

100. Which keeps you SAFEST during a thunderstorm?

- (A) swimming in a pool
- (B) holding a metal-tipped umbrella up high
- (C) staying inside a building
- (D) standing under a tall lone tree